Arpon Kapuria

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Summary

Building end-to-end AI/ML systems across LLMs, RAG and Agentic AI — from experimentation and evaluation to scalable inference, serving, and deployment.

Education

National Institute of Technology Tiruchirappalli — **Tiruchirappalli, India**
Bachelor of Technology (*Computer Science and Engineering*) — November 2020 – June 2024

Skills

- **Programming Languages:** Python, C, C++, SQL, Bash
- **AI / ML:** PyTorch, Hugging Face Transformers, PEFT, TRL, Unsloth, LangChain, LangGraph, vLLM
- **Backend & Data:** FastAPI, Ray, PostgreSQL, Redis, Vector Databases
- **Infrastructure & MLOps:** MLflow, DVC, Docker, Kubernetes, Terraform, AWS, Git, CI/CD
- **Observability:** LangSmith, Prometheus, Grafana, OpenTelemetry

Work Experience

Advanced Machine Intelligence Research Lab — **Dhaka, Bangladesh**
Research Assistant | *Advisor: Prof. M. Firoz Mridha* — November 2025 – Present

- Led the research design of a memory- and alignment-based framework for trustworthy mental health conversational agents, with an evaluation protocol for safety, empathy and temporal consistency.

Indian Institute of Technology Bombay — **Mumbai, India**
Research Intern | MeDAL Lab | *Advisor: Prof. Amit Sethi* — May 2023 – July 2023

- Conducted an in-depth survey on self-supervised learning Frameworks and reproduced BYOL and simCLR in **PyTorch**. Experimented with loss functions and architectural variations to study their impact on learned representations.
- Implemented a DICOM anonymization pipeline using **Pydicom** to remove patient metadata from **5,000+ scans** and integrated CRAFT model to detect text regions in X-ray images for masking PHI.

National Institute of Technology Tiruchirappalli — **Tiruchirappalli, India**
Undergraduate R&D Assistant | Industrial Automation Lab — October 2022 – February 2023
Project: Assam Smart Home Automation — *Advisors: Prof. M. Brindha, Prof. G. S. Ilango*

- Developed a cross-platform application to automate energy operations and digital billing for a solar-powered microgrid in Assam, India, serving **20+ households**, as part of a govt. funded project.
- Integrated **Flutter** frontend with Django-based REST API and deployed an MQTT broker enabling real-time data exchange with **50+ IoT sensors** for energy consumption and control.

Project

Agentic RAG Pipeline with Hybrid Retrieval & Scalable Deployment — March 2026 – April 2026

- Designed an end-to-end Agentic RAG system on with a **LangGraph** planner routing queries across hybrid retrieval (**Qdrant** vector search and **Neo4j** graph traversal), using HyDE, query rewriting, and a cross-encoder reranker to improve multi-hop reasoning beyond embedding-only pipelines.
- Deployed LLM inference via **vLLM** on **Ray Serve** with AWQ quantization, **continuous batching**, and **tensor parallelism**; served BGE-M3 embeddings with **fractional GPU allocation** and **torch.compile**, scaling to **128 concurrent sequences per replica** under autoscaled GPU worker groups.
- Provisioned multi-AZ AWS infrastructure with **Terraform** (**EKS**, **Aurora Serverless v2**, **ElastiCache HA**, **S3** with lifecycle tiering) and configured **Karpenter** with separate CPU and GPU provisioners — **30-second GPU scale-to-zero** and **spot consolidation** — reducing idle compute cost without compromising autoscaling responsiveness.
- Instrumented the full request path with **OpenTelemetry** traces, **Prometheus** metrics, and **Grafana** dashboards; evaluated retrieval and generation quality offline via **RAGAS** and an **LLM-as-a-Judge** pipeline against a curated golden dataset.